PS#3: Integration by substitution and integration by parts - Answer key

1. There are some integrals you can do in multiple different ways. Here's one for you to think about.

   (a) Compute $\int \cos(x)\sin(x)\,dx$ using substitution.

   I think I'm going to say $u = \sin(x)$. Then $\frac{du}{dx} = \cos(x)$, so $du = \cos(x)\,dx$. Well how about that, I have $\cos(x)\,dx$ in the integrand!

   $$\int \cos(x)\sin(x)\,dx = \int \underbrace{\sin(x)}_{u}\underbrace{\cos(x)\,dx}_{du} = \int u\,du = \frac{1}{2}u^2 + C = \frac{1}{2}(\sin(x))^2 + C$$

   (Notice that we could also have used $u = \cos(x)$, $du = -\sin(x)\,dx$. Do we get a different answer that way, or is it really the same answer in disguise?)

   (b) Compute $\int \cos(x)\sin(x)\,dx$ using integration by parts.

   I think I'll use this combination of u and dv:

   $$\begin{array}{ll} u = \sin(x) & \\ dv = \cos(x)\,dx & \end{array} \quad \text{so then} \quad \begin{array}{ll} u = \sin(x) & v = \sin(x) \\ du = \cos(x)\,dx & dv = \cos(x)\,dx \end{array}$$

   Then using the integration by parts formula,

   $$\int \cos(x)\sin(x)\,dx = \sin(x)\sin(x) - \int \sin(x)\cos(x)\,dx$$

   Oh look, the second integral is the same as the first! Let's add it back to the other side.

   $$2\int \cos(x)\sin(x)\,dx = \sin^2(x)$$
   $$\int \cos(x)\sin(x)\,dx = \frac{1}{2}\sin^2(x) + C$$

   (Notice that this probably would have worked if we did $u = \cos(x)$ and $dv = \sin(x)\,dx$. Do we get a different answer that way, or is it really the same answer in disguise?)

2. (Section 5.4 Exercise 7) For each of the following indefinite integrals, determine whether you would use u-substitution, integration by parts, neither, or both to evaluate the integral. In each case, write one sentence to explain your reasoning, and include a statement of any substitutions used. Finally, use your chosen approach to evaluate each integral.

   (a) $\int x^2\cos(x^3)\,dx$

   This one is pretty clearly substitution, because the x^2 looks like the derivative of the x^3 filling. Indeed, if $u = x^3$, then $\frac{du}{dx} = 3x^2$, so $dx = \frac{du}{3x^2}$.

   $$\int x^2\cos(x^3)\,dx = \int x^2\cos(u)\,\frac{du}{3x^2} = \frac{1}{3}\int \cos(u)\,du = \frac{1}{3}\sin(u) + C = \frac{1}{3}\sin(x^3) + C$$

(b) $\int x^5 \cos(x^3)\,dx = \int x^3 \cdot (x^2 \cos(x^3))\,dx$

This one we'll use integration by parts, specifically because we just figured out how to integrate $x^2\cos(x^3)$.

$$u = x^3 \qquad dv = x^2\cos(x^3)\,dx \quad \text{so then} \quad \begin{array}{ll} u = x^3 & v = \frac{1}{3}\sin(x^3) \\ du = 3x^2\,dx & dv = x^2\cos(x^3)\,dx \end{array}$$

Using the integration by parts formula:

$$\begin{aligned} \int x^3 \cdot (x^2\cos(x^3))\,dx &= x^3 \cdot \tfrac{1}{3}\sin(x^3) - \int \tfrac{1}{3}\sin(x^3)\cdot 3x^2\,dx \\ &= \frac{1}{3}x^3\sin(x^3) - \int \sin(x^3)\cdot x^2\,dx \end{aligned}$$

We'll use the substitution $t = x^3$, $dt = 3x^2\,dx$ to evaluate this second integral.

$$\begin{aligned} &= \frac{1}{3}x^3\sin(x^3) - \int \sin(t)\,\frac{dt}{3} \\ &= \frac{1}{3}x^3\sin(x^3) + \frac{1}{3}\cos(t) = \frac{1}{3}x^3\sin(x^3) + \frac{1}{3}\cos(x^3) + C \end{aligned}$$

(c) $\int x\ln(x^2)\,dx$

This one looks like substitution to me, because the x on the outside looks like the derivative of the x^2. Let's say $u = x^2$, so $\frac{du}{dx} = 2x$, so $dx = \frac{du}{2x}$.

$$\begin{aligned} \int x\ln(x^2)\,dx &= \int x\ln(u)\,\frac{du}{2x} = \frac{1}{2}\int \ln(u)\,du - \text{We figured out that integral in class!} \\ &= \frac{1}{2}(u\ln(u) - u) = \frac{1}{2}\left(x^2\ln(x^2) - x^2\right) + C \end{aligned}$$

An alternative strategy using integration by parts: logarithmic functions are often a really nice choice for u because they get so much simpler when you differentiate them.

$$u = \ln(x^2) \qquad dv = x\,dx \quad \text{so} \quad \begin{array}{ll} u = \ln(x^2) & v = \frac{1}{2}x^2 \\ du = \frac{1}{x^2}\cdot 2x\,dx & dv = x\,dx \end{array}$$

Applying the integration by parts formula:

$$\begin{aligned} \int x\ln(x^2)\,dx &= \ln(x^2)\cdot\tfrac{1}{2}x^2 - \int \frac{1}{2}x^2\cdot\frac{1}{x^2}\cdot 2x\,dx \\ &= \ln(x^2)\cdot\tfrac{1}{2}x^2 - \int x\,dx\ (!!) \\ &= \frac{1}{2}x^2\ln(x^2) - \frac{1}{2}x^2 + C \end{aligned}$$

(d) $\int \sin(x^4)\,dx$

This function has no elementary antiderivative! u-substitution won't work because you can't make up your du; parts won't work either.

(e) $\int x^3\sin(x^4)\,dx$

Ah ha, now we can use substitution. If $u = x^4$, then $\frac{du}{dx} = 4x^3$, so $dx = \frac{du}{4x^3}$.

$$\int x^3\sin(x^4)\,dx = \int x^3\sin(u)\,\frac{du}{4x^3} = \frac{1}{4}\int \sin(u)\,du = \frac{1}{4}(\cos(u)) = -\frac{1}{4}\cos(x^4) + C$$

(f) $\int x^7\sin(x^4)\,dx = \int x^4\cdot(x^3\sin(x^4))\,dx$

We'll use a similar trick as we did in part (b):

$$\begin{array}{ll} u = x^4 & \\ dv = x^3\sin(x^4)\,dx & \end{array} \quad \text{so then} \quad \begin{array}{ll} u = x^4 & v = -\frac{1}{4}\cos(x^4) \\ du = 4x^3\,dx & dv = x^3\sin(x^4)\,dx \end{array}$$

Applying integration by parts:

$$\begin{aligned}\int x^4\cdot(x^3\sin(x^4))\,dx &= x^4\cdot\left(-\tfrac{1}{4}\cos(x^4)\right) - \int -\tfrac{1}{4}\cos(x^4)\cdot 4x^3\,dx \\ &= -\tfrac{1}{4}x^4\cos(x^4) + \int x^3\cos(x^4)\,dx\end{aligned}$$

We'll use the substitution $t = x^4$, $dt = 4x^3\,dx$ to evaluate the second integral.

$$= -\frac{1}{4}x^4\cos(x^4) + \frac{1}{4}\int \cos(t)\,dt = -\frac{1}{4}x^4\cos(x^4) + \frac{1}{4}\sin(t) = -\frac{1}{4}x^4\cos(x^4) + \frac{1}{4}\sin(x^4$$

3. You're working on some problems with your friend Leonhard, who writes down the following:

$$\int (4t+3)\sin(2t)\,dt$$

$$\begin{array}{ll} u = \sin(2t) & v = 2t^2 + 3t \\ du = 2\cos(2t)\,dt & dv = (4t+3)\,dt \end{array}$$

$$\int (4t+3)\sin(2t)\,dt = \sin(2t)(2t^2+3t) - \int \left[(2t^2+3t)\cdot 2\cos(2t)\right]\,dt$$

Leonhard is now stuck: he's not sure how to attack this new integral. Can you help him get unstuck and find this integral correctly?

Leonhard's issue is that he made a bad choice for u and dv. In particular, the thing he chose for dv got more complicated (went up by a power of t) when he integrated it. He can solve his issue by switching his choice:

$$\begin{array}{ll} u = 4t+3 & v = -\frac{1}{2}\cos(2t) \\ du = 4\,dt & dv = \sin(2t)\,dt \end{array}$$

Then the integration by parts formula works out much nicer (look how much simpler du is!):

$$\int (4t+3)\sin(2t)\,dt = (4t+3)\cdot\tfrac{-1}{2}\cos(2t) - \int \tfrac{-1}{2}\cos(2t)\cdot 4\,dt = -(4t+3)\cdot\tfrac{1}{2}\cos(2t) + 2\int \cos(2t)\,dt$$
$$= -(4t+3)\cdot\tfrac{1}{2}\cos(2t) + \sin(2t) + C$$